

Deliverable 1: Requirement Tables

Name: Prathamesh Kumar Singh **SRN:** PES1UG24CS341

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.	High	Pass: Voucher code generated with cryptographic checksum. Fail: Expired or duplicate voucher redeemed.	Core functionality enabling cross-brand loyalty point consolidation and asset liquidity.
FR-002	Functional	The system shall query partner merchant APIs in real time to fetch dynamic conversion rates before confirming any point exchange.	High	Pass: Token conversion recalculates dynamically reflecting current merchant rate. Fail: Point conversion processes with stale (> 5 min) or missing exchange rates.	Eliminates cross-merchant currency valuation arbitrage and protects wallet token reserves.
FR-003	Functional	The system shall place an atomic lock on selected digital gift card vouchers during active checkout sessions.	High	Pass: Voucher status transitions to LOCKED with a 10-minute hold window.	Guarantees transactional consistency and prevents double-spending in high-concurrency environments.

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				Fail: Duplicate checkout sessions simultaneously lock or redeem the same voucher.	
FR-004	Functional	The system shall generate daily batch reconciliation summaries and settle universal exchange credits for merchant partners.	Medium	Pass: End-of-day ledger balances balance to zero across all verified transactions. Fail: Discrepancies between issued vouchers and deducted tokens exceed zero drift.	Provides auditable financial settlement and operational transparency for participating partners.
FR-005	Functional	The system shall prompt for Multi-Factor Authentication (MFA) on all point conversions or redemptions exceeding 5,000 exchange credits.	Medium	Pass: Transaction proceeds only upon successful OTP/biometric validation. Fail: High-value voucher is issued without secondary factor authorization.	Mitigates unauthorized point draining in the event of compromised primary account credentials.

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NFR-001	Non-Functional	Point conversion and gift card redemption operations must process within 250 ms with zero ledger balance drift.	High	Pass: Response latency $\leq$ 250 ms under peak load with 100% ledger consistency. Fail: Response time $>$ 250 ms or unverified balance drift occurs.	Maintains real-time interactive responsiveness and absolute financial accounting accuracy.
NFR-002	Non-Functional	The system shall encrypt all stored voucher secrets, cryptographic checksums, and in-transit payloads using AES-256 and TLS 1.3.	High	Pass: All database logs and network traffic show zero plaintext token/voucher payloads. Fail: Plaintext transmission or storage detected during compliance audit.	Enforces strict security compliance and prevents unauthorized interception of digital assets.